IT4010 Research Project – 2025 July Topic Assessment Form

Project ID: 25-26J-062

1. Topic (12 words max)

InvisiGuard - Intelligent System for Predicting Cyber Threats Real Time

2. Research group the project belongs to

IAS - Information Assurance & Security

3. Specialization of the project belongs to

Cyber Security (CS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

InvisiGuard - Intelligent System for Predicting Cyber Threats Real Time

The increasing sophistication of cyberattacks, including both known and zero-day attacks, presents a significant challenge to traditional cybersecurity systems, which rely on predefined signatures and rule-based detection methods. Known attacks can often be detected using signature-based approaches, but zero-day attacks, which exploit previously unknown vulnerabilities, remain elusive. Additionally, existing systems fail to proactively predict potential attack scenarios, leaving organizations vulnerable to emerging threats.

Predicting Known Attacks Using System Call Patterns

Known attacks are often detected using signature-based detection systems that rely on predefined attack patterns. However, these systems face limitations in terms of accuracy and timeliness, especially as new variants of known attacks emerge or when attack strategies evolve. System calls patterns, which provide a detailed trace of how a program interacts with the operating system, hold the potential to offer more dynamic and effective methods for detecting these known attacks. Despite their importance, current systems often fail to fully leverage system call data for real-time attack detection.

Predicting Known Insider Attacks Using User Behavioral Patterns

Detecting known attacks based on user behavior presents a significant challenge due to the complexity and variability of legitimate user activities. Traditional security systems primarily rely on static rules or signature-based methods, which often fail to capture the dynamic nature of user interactions with systems. As user behavior evolves and adapts over time, especially in response to changing work environments, it becomes increasingly difficult to identify deviations that may signal a malicious attack. This research aims to bridge this gap by using machine learning to analyze user behavioral patterns and predict known attacks. The problem lies in the ability to accurately distinguish between benign user behavior and potentially harmful activities, even as legitimate actions change over time.

Predicting Zero-Day Attacks Using Network Traffic Patterns

Zero-day attacks exploit vulnerabilities in a system that are unknown to security vendors, making it extremely difficult to detect using traditional signature-based methods. Network traffic patterns hold valuable insights into the activities of both legitimate and malicious users, but existing systems often fail to detect zero-day attacks due to the subtlety of these attacks and the inability to model complex, evolving traffic patterns.

Predicting Zero-Day Attacks Using Endpoint Log Patterns

Zero-day attacks, which exploit previously unknown vulnerabilities, are particularly challenging to detect at the endpoint level due to their stealthy nature. Traditional endpoint security systems rely on known attack signatures or heuristic approaches, which often fail to detect these novel, undetected threats. Endpoint logs, which record user actions, system processes, and application interactions, contain valuable information that could reveal the presence of zero-day attacks.

References:

An intrusion detection model to detect zero-day attacks in unseen data using machine learning : https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0308469#:~:text=The%20rapid%20evolution%20of%20technology%20has%20led%20to,the%20CIC-MalMem-2022%20dataset%20and%20autoencoders%20for%20anomaly%20detection.

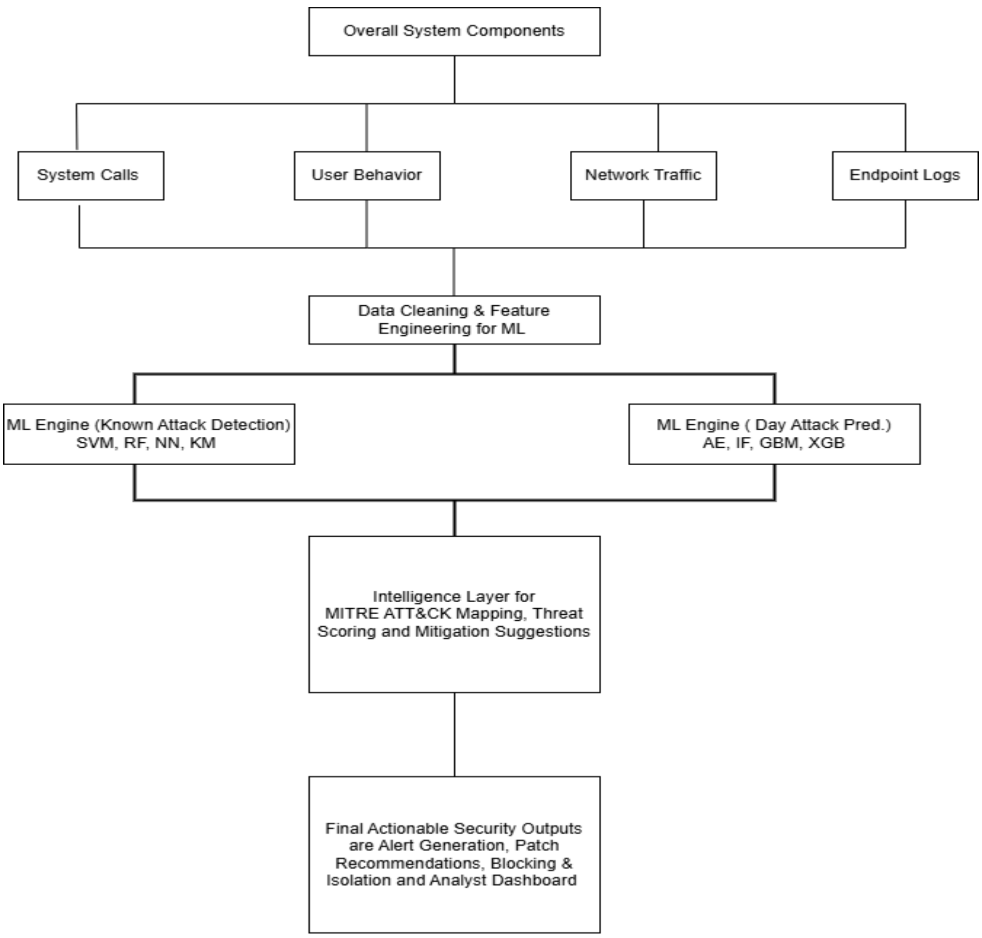
An Analysis of Signature-Based Components in Hybrid Intrusion Detection Systems: https://ieeexplore.ieee.org/document/10449209

Bridging the Gap: A Pragmatic Approach to Generating Insider Threat Data: Bridging the Gap: A Pragmatic Approach to Generating Insider Threat Data | IEEE Conference Publication | IEEE Xplore

Analyzing Data Granularity Levels for Insider Threat Detection Using Machine Learning: https://ieeexplore.ieee.org/document/8962316

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Our solution, InvisiGuard is a next-generation, machine learning-based cybersecurity system designed to predict and prevent both known and zero-day attacks. Unlike traditional signature-based systems that react after threats are identified, this solution adopts a proactive, intelligence-driven approach. It integrates multiple machine learning models to analyze diverse data sources such as system call patterns, user behavioral logs, network traffic flows and endpoint event logs. By combining supervised learning techniques (Support Vector Machines, Random Forest, XGBoost) for detecting known threats with unsupervised learning techniques (Autoencoders, Isolation Forest) for identifying previously unseen anomalies, the system provides a comprehensive and adaptive defense mechanism. Additionally, it incorporates a centralized intelligence layer that correlates prediction results with frameworks like MITRE ATT&CK to generate real time, actionable insights, including alerting, patching suggestions and automated threat mitigation strategies. This integrated, multi layered architecture transforms cybersecurity from a reactive model into a proactive and resilient security posture capable of defending against evolving digital threats.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project contains four main components to full fill the final product. Which are System Calls Analysis, User Behavior Monitoring, Endpoint Log Inspection, Network Traffic Analysis. These components going to cover various types of domain topics including:

1.Cybersecurity / Threat Detection Expertise

2.Operating System Internals & System Programming

3.Machine Learning / Data Science

4.Log Analysis & Endpoint Monitoring

5.Network Security / Packet Analysis

6.Software Development / Product Engineering

7.DevOps / Deployment

Data for generating intelligence measures will be collected from various Security Intelligence sources:

1.MITRE Attack Framework: Provides structured information about different types of attacks, their lifecycle, and potential vulnerabilities.

2.Open Threat Intelligence Platforms: Sources like MITRE ATT&CK, CVE databases, and Virus Total to gather information about attack techniques, tactics, and reported vulnerabilities.

3.Security Incident Data: Real-world incident reports, attack logs, and threat intelligence feeds to understand the patterns and behaviors of known and zero-day threats.

Component 1 datasets

- **AWSCTD (Windows Syscall Traces)** - https://fkie-cad.github.io/COMIDDS/content/datasets/awsctd/
- **Malware API Call Sequences (Windows)** - https://ieee-dataport.org/open-access/malware-api-call-dataset
- **SysCall Dataset (UAV Usage)** - https://data.mendeley.com/datasets/vfvw7g8s8h/1
- **The CICIDS 2017 dataset** - https://www.unb.ca/cic/datasets/ids-2017.html

Component 2 datasets:

- **Intrusion Detection Dataset - https://www.kaggle.com/datasets/dnkumars/cybersecurity-intrusion-detection-dataset**
- **Insider threat detection - https://www.kaggle.com/datasets/mrajaxnp/cert-insider-threat-detection-research**

Component 3 datasets:

- **Dataset for anomaly and intrusion detection** - https://www.kaggle.com/datasets/galaxyh/kdd-cup-1999-data
- **Realistic modern network traffic -** https://research.unsw.edu.au/projects/unsw-nb15-dataset
- **Zero-Day Anomaly Detection -** https://www.unb.ca/cic/datasets/ids-2018.html

Component 4 datasets:

- **Dataset for Sysmon log -** **https://www.kaggle.com/datasets/surapitthaibkhuang/malware-eventid-time-series-dataset-by-sysmon-log** **https://www.kaggle.com/datasets/ramachandrabharadwaj/sysmon**
- **Dataset for Registry key – https://raw.githubusercontent.com/OTRF/Security-Datasets/master/datasets/atomic/windows/defense_evasion/host/empire_enable_rdp.tar.gz**

8. Objectives and Novelty

The novelty of this research lies in its integrated machine learning framework that predicts both known and zero-day attacks through the analysis of multiple data sources, including system calls, user behavior, network traffic and endpoint logs. Unlike traditional systems that focus on specific attack types or predefined signatures, this approach combines multiple predictive models to offer real-time, intelligence-driven security insights.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Nawarathna.N.M.I.M IT22343116	Known Attack Prediction by Analyzing System Call Patterns.	• Collect and analyze system call data from systems under attack. • Classify the attack patterns using ML algorithms. • Improve prediction accuracy for known attack types. • Develop a real-time system that continuously monitors system calls, detects known attacks instantly, and provides actionable insights for prompt mitigation.	Since there are many software solutions which using ML and system call sequences to identify attacks, we are using a "Two stage (Hybrid)" system for this component. Stage 1 (Fast filter) - Use Random Forest or SVM. Stage 2 (Deep analyzer) - Use RNN/LSTM only on suspicious sequences. Also, we are collecting not just basic syscall features but also; • Arguments. (e.g., file paths, IPs)

			• Return values. (e.g., success/failure) • Timing between calls. • Parent-child process links.
Pallewaththa P.A IT22133304	Predicting Known Insider Attacks Using User Behavioral Patterns	• Collect and preprocess user behavior data • Extract key behavioral features (login frequency, file access, application usage, network interactions) • Collecting web browsing history to detect anomaly detections • Established a baseline model of typical user behavior to compare new actions against and identify anomalies indicative of known threats. • Train ML models using unsupervised learning with Random Forest • Implemented the trained models to identify and classify abnormal behaviors	The novelty of this research lies in the traditional methods that focus primarily on system-level or signature-based analysis. This study offers using Behavioral Graph Construction and Analysis can improve the accuracy of identifying known insider attacks. analyze user activity data such as login patterns, file access, and network communication behaviors to create an evolving behavioral model. Machine learning models will be trained to recognize anomalous behavior, helping to predict known attacks based on these deviations.

		as potential known insider attack indicators.	
Nawarathna.N.P.D.T IT22117496	Predicting Zero-Day Attacks Using Network Traffic Patterns	• Collect and preprocess network traffic data, focusing on packet-level details and flow statistics. • Identify anomalous network traffic patterns that could indicate potential zero-day attacks. • Apply unsupervised machine learning techniques to detect deviations from normal traffic behavior. • Enhance the predictive accuracy of zero-day attack detection through anomaly detection methods. • Provide a scalable and adaptive solution for real-time zero-day attack prediction.	The novelty is applying advanced unsupervised dynamic machine learning techniques such as Autoencoders and Isolation Forest to predict zero-day attacks based on detailed analysis of network traffic patterns, rather than relying on predefined signatures. Many studies have addressed signature-based zero-day attack prediction using network patterns. However, only a limited number of studies have focused on predicting zero-day attacks solely through network traffic analysis. Among those, most have produced false positives when attempting to detect previously unseen attacks. By employing machine learning techniques like Autoencoders and Isolation

			Forest, this approach aims to reduce false positives while increasing detection accuracy to nearly 99%. By focusing on traffic anomalies instead of predefined attack signatures, these methods offer a more flexible and adaptive way to detect previously unknown threats. The integration of these advanced techniques provides an innovative and more accurate real-time prediction of zero-day attacks, reducing the vulnerability window and enhancing overall network security.
Nethkalum G M H IT22562906	Zero-Day Attack Prediction by Analyzing Endpoint Log Patterns	• Extract system event log features such as process creation, system errors, and application crashes that may reflect abnormal behaviors. • Ingest and normalize raw Sysmon logs and other relevant endpoint telemetry, focusing on	This novelty method for detecting zero-day attacks through advanced endpoint log analysis. By extracting granular details from system event logs, such as process creation and application crashes, it identifies abnormal behaviors that signal potential threats. The

		extracting granular details like full command-line arguments, file hashes, and specific registry key values. • Track file integrity changes to detect unauthorized modification • Analyze application execution patterns by recording the frequency and order of executed program • Using unsupervised models like Isolation Forests to develop a high-accuracy model.	system normalizes raw Sysmon logs, focusing on full command-line arguments and registry key values, and monitors file integrity for unauthorized changes. Using unsupervised models like Isolation Forests, it achieves high-accuracy anomaly detection, enabling the identification of zero-day threats without relying on existing threat signatures, providing a proactive defense against emerging cyber threats.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Nawrathna N.M.IM IT22343116	This component will cover how the system calls can be useful to identify malware in a system. The System Call Pattern Analysis component reflects core cybersecurity principles by focusing on detecting malicious behavior at the operating system level where most real-world attacks originate. System calls serve as the interface between user applications and critical OS resources like files, memory, and networks. By monitoring and analyzing patterns of these calls, the component can identify signs of known attacks such as privilege escalation, file tampering, or process injection.
Pallewaththa P.A IT22133304	Predicting Known Insider Attacks Using User Behavioral Patterns, supports Cybersecurity by enhancing threat detection through user behavior analysis. It focuses on identifying known attacks using real-time user activity patterns, which aligns with intrusion detection, behavioral monitoring, and incident response all core areas in cybersecurity. This approach strengthens proactive defense mechanisms, moving beyond traditional signature-based methods and contributing to modern adaptive security systems.
Nawarathna N.P.D.T IT22117496	This focuses on detecting zero-day attacks, one of the most critical and evasive threats in cybersecurity. By leveraging machine learning to analyze network traffic anomalies, it reflects a modern, proactive approach to threat detection beyond traditional signature based systems. The use of packet level and flow level data for building intelligent models demonstrates practical skills in traffic analysis, anomaly detection, and AI driven defense mechanisms. My component enhances core cybersecurity competencies in intrusion detection, threat intelligence and real-time network security monitoring, which are all fundamental to advanced cybersecurity practices.
Nethkalum G.M.H IT22562906	The component focuses on proactively predicting zero-day attacks using endpoint log analysis. By applying advanced machine learning models such as XGBoost and Gradient Boosting Machines, it moves beyond traditional signature-based defenses and toward intelligent, adaptive threat detection. The use of detailed endpoint data such as process creation, user activity, file integrity, and system errors demonstrate practical skills in log analysis, anomaly detection, and behavioral modeling.

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Amila	Nuwan Senarathne	
Co-Supervisor	Mr.	Deemantha	Siriwardana	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				